Toward PK-Free Molecular Hamiltonians: Two-Center Sturmian Integrals and Coupled Composition*

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The composed natural geometry framework (Paper 17) constructs molecular Hamiltonians by partitioning electrons into single-center blocks and coupling them via Phillips–Kleinman (PK) pseudopotentials. This produces structurally sparse qubit Hamiltonians—a Pauli count exactly linear in the qubit count across molecules at fixed basis, $N_{\text{Pauli}} = 27.90 \times Q$, with 54×–317× fewer terms than equal-qubit Gaussian baselines (Paper 14; the retired pair-diagonal rule gave $O(Q^{2.5})$ and 51–1,712×)—but PK limits accuracy to 5.3–19.4% equilibrium bond length error. PK has resisted improvement: six modification attempts have failed (v2.0.6–23), and the overcounting diverges at higher angular momentum.

We report a negative result (Track CB, v2.0.37): replacing PK with explicit cross-block electron-electron integrals *without* corresponding cross-center nuclear attraction integrals produces an unbalanced Hamiltonian (29% error, worse than PK's 15%). The root cause is identified: two-center one-body integrals $\langle \psi^A_{nlm} | V_B | \psi^A_{n'l'm'} \rangle$, where an orbital on center A feels the nuclear potential of center B, are required for energy balance but absent from the single-center framework.

We observe that these missing integrals are closely related to the Shibuya–Wulfman integrals of the Coulomb Sturmian literature [7, 10, 12], which are computed via Fock's momentum-space projection onto S^3—the same conformal map that underlies the GeoVac framework (Paper 7). The mathematical foundation is therefore already in place. We review the convergence properties of the Shibuya–Wulfman expansion, identify the Coulomb resolution technique [11] as a potentially faster alternative, and outline the "coupled composition" architecture that would result from incorporating two-center integrals. If the angular expansion converges in few terms at molecular bond lengths, the coupled Hamiltonian would be PK-free, structurally sparse, and significantly more accurate than the current composed framework. This paper presents the balanced coupled architecture, documents its mathematical foundations, and reports computational benchmarks across three molecules (LiH, BeH_2, H_2O).

We report computational results from the first three phases. The convergence concern is resolved: the multipole expansion of the cross-center potential terminates exactly at $L_{\max} = 2l_{\max}$ by Gaunt selection rules (33 one-body terms at $n_{\max} = 2$). The balanced Hamiltonian has 2,726 Pauli terms (Conjecture 1 holds in its ratio form only; see Sec. VII F). Four-electron FCI gives R_{eq} error of 7.0% and $D_e = 0.037$ Ha—the only bound configuration among all four coupling schemes tested (Conjecture 2 not confirmed in strict sense, but the comparison crosses FCI regimes). At $n_{\max} = 3$ ($Q = 84$), single-point FCI gives 0.20% energy error (down from 1.8%) with analytical integrals (incomplete gamma functions, machine precision), confirming convergence through $n_{\max} = 3$; at $n_{\max} = 4$ the model crosses *past* exact (over-binding), so the fixed-geometry error is not monotonically improvable either. The R_{eq} drift is structural to the block decomposition (+0.057 then +0.023 bohr at the two available steps— decelerating, not a constant rate; 3× smaller than PK's +0.15–0.22 bohr/$l_{\max}$), not numerical.

I. INTRODUCTION: THE PK PROBLEM

The composed natural geometry framework [5] partitions electrons into groups—core, bond pairs, lone pairs—each inhabiting its own natural coordinate system. Inter-group coupling is approximated by the Phillips–Kleinman (PK) pseudopotential, a one-body operator that provides core-valence orthogonality and an effective repulsive barrier.

PK is the framework's accuracy bottleneck. At $l_{\max} = 2$ (the optimal operating point), LiH achieves 5.3% R_{eq} error; BeH_2 gives 11.7%; H_2O gives 19.4%. These errors are structural, not numerical: the $l_{\max}$ divergence (+0.15–0.22 bohr per additional angular channel) is confirmed to originate in PK overcounting, not in the adiabatic approximation (Track BQ, v2.0.32; sprint and track designations are anchors into the project's internal chronicle, `CHANGELOG.md` in the repository). Six attempts to modify PK have failed (projector, spectral, self-consistent, R-dependent, l-dependent, l-dependent on 2D solver), and the ab initio R_{ref} derivation failed because the relevant scale (∼3 bohr) is molecular, not atomic (Track AD, v2.0.18).

At the same time, PK is essential for the composed framework's sparsity: it makes the electron repulsion integral (ERI) tensor block-diagonal, yielding a Pauli count exactly linear in the qubit count across molecules at fixed basis ($N_{\text{Pauli}} = 27.90 \times Q$, Paper 14) [4]. Without PK, the block decomposition has no coupling at all.

The question motivating this paper is: *can PK be re-*

* Paper 19 in the Geometric Vacuum series

placed by explicit inter-block integrals that preserve structural sparsity while eliminating the accuracy ceiling?

II. THE NEGATIVE RESULT: COUPLED COMPOSITION WITHOUT V_{ne}

Track CB (v2.0.37) tested the simplest version of this idea: remove PK from the composed LiH Hamiltonian and add explicit cross-block two-electron integrals $\langle pq|rs\rangle$ where p, q belong to one block and r, s to another.

The cross-block ERIs were computed using existing infrastructure (`cross_block_mp2.py`, originally built for Track BX-3b). Key census results for LiH at $n_{\max} = 2$, $Q = 30$:

TABLE I. Cross-block ERI census for LiH at $n_{\max} = 2$.

Metric	Within-block	Cross-block
Nonzero ERIs	195	130
Max $\lvert\langle pq\|rs\rangle\rvert$ (Ha)	—	0.891
Mean $\lvert\langle pq\|rs\rangle\rvert$ (Ha)	—	0.142

The cross-block/within-block ERI ratio is 2/3, consistent with the selection-rule counting bound of $\leq 2\times$ from Paper 14, Sec. V.E. Gaunt selection rules ($|\Delta l| \leq 1$, $|\Delta m| \leq 1$) are preserved across block boundaries.

The resulting qubit Hamiltonian has 2,702 Pauli terms (3.22× the composed value of 838) and non-identity 1-norm 80.5 Ha (2.10× the composed value of 38.3 Ha). However, the FCI energy tells the real story:

TABLE II. FCI accuracy for LiH at $n_{\max} = 2$ under different coupling schemes. Exact: −8.071 Ha.

Configuration	E_{FCI} (Ha)	Error
No PK, no cross-block (decoupled)	−7.195	10.9%
Composed (PK, no cross-block)	−6.863	15.0%
Coupled (no PK, with cross-block V_{ee})	−5.734	29.0%

The coupled Hamiltonian is *worse* than both alternatives. The root cause is clear: cross-block ERIs add inter-group electron-electron *repulsion* without the balancing cross-center electron-nucleus *attraction*. The Hamiltonian is energetically unbalanced.

Observation 1. *The decoupled configuration (no PK, no cross-block interactions) gives 10.9% error—better than PK's 15.0%. This implies that PK overcorrects at* $n_{\max} = 2$*: the blocks are more accurate when left independent than when coupled via PK.*

This observation is consistent with the $l_{\max}$ divergence diagnostic: PK's repulsive barrier systematically overcounts the core-valence interaction energy, and the overcounting worsens with basis size.

III. THE MISSING PIECE: TWO-CENTER NUCLEAR ATTRACTION

The coupled Hamiltonian of Sec. II is incomplete. A physically balanced molecular Hamiltonian requires, for each electron:

1. Kinetic energy (within-block, already present).
2. Same-center nuclear attraction $V_A = -Z_A/r_A$ (within-block, already present).
3. Cross-center nuclear attraction $V_B = -Z_B/r_B$, where the electron on center A feels the potential of nucleus B (missing).
4. Electron-electron repulsion, both within-block (present) and cross-block (added in Track CB).

Item 3 requires two-center one-body integrals of the form

$$I^{AB}_{nlm,n'l'm'} = \left\langle \psi^A_{nlm} \left| \frac{-Z_B}{|\mathbf{r} - \mathbf{R}_B|} \right| \psi^A_{n'l'm'} \right\rangle \tag{1}$$

where ψ^A_{nlm} is a hydrogenic orbital centered on nucleus A, and the potential is from nucleus B at position $\mathbf{R}_B$. These integrals are standard in the Slater-type orbital (STO) literature but have never been computed in the GeoVac framework, which has until now been strictly single-center at each level.

IV. CONNECTION TO COULOMB STURMIAN THEORY

GeoVac's hydrogenic orbitals are Coulomb Sturmians with the identification $k = Z/n$, where k is the Sturmian exponent. This is not merely a notational correspondence: the entire GeoVac framework—the graph Laplacian on S^3 (Paper 7), the Fock projection, the angular momentum selection rules—is built on the same mathematical structure that the Avery group has developed for molecular Coulomb Sturmian calculations since the 1990s [9, 10, 12, 13].

A. Shibuya–Wulfman integrals

When a Coulomb Sturmian centered on A is expanded in the basis centered on B, the expansion coefficients are the Shibuya–Wulfman integrals [7]:

$$S_{\tau',\tau} = \int d^3\mathbf{r}\, \chi^*_{\tau'}(\mathbf{r} - \mathbf{R}_B)\, \frac{k}{r_A}\, \chi_\tau(\mathbf{r} - \mathbf{R}_A) \tag{2}$$

where χ_τ denotes a Coulomb Sturmian with quantum numbers $\tau = (n, l, m)$, and the weighting k/r_A arises from the potential-weighted orthogonality of the Sturmian basis.

In Fock's momentum-space representation, Coulomb Sturmians map to hyperspherical harmonics on S^3, and Shibuya–Wulfman integrals become overlap integrals of translated hyperspherical harmonics. The translation is parameterized by

$$S \equiv k \cdot R_{AB} \tag{3}$$

where $R_{AB} = |\mathbf{R}_A - \mathbf{R}_B|$ is the internuclear distance and k is the common Sturmian exponent.

The nuclear attraction integral of Eq. (1) is closely related to the Shibuya–Wulfman integral: both involve a Coulomb potential centered on a different nucleus acting on an orbital. Koga and Matsuhashi [8] derived sum rules for nuclear attraction integrals over hydrogenic orbitals that express them as finite sums over Clebsch–Gordan coefficients—the same angular momentum algebra that generates Gaunt integrals in the GeoVac framework.

Heterogeneous exponents. The standard Shibuya–Wulfman formalism assumes a common Sturmian exponent k across all centers. GeoVac's composed blocks use different effective charges Z_{eff} per block, giving different exponents $k_b = Z_{\text{eff},b}/n$ for each block b. The generalized overlap integrals between Sturmians of different exponents exist in the literature; Koga and Matsuhashi [8] give sum rules for the common-exponent case, and the convergence properties for heterogeneous exponents may differ from the common-k case. Any implementation must use these generalized integrals, and the convergence diagnostic (Sec. VIII) must be performed with the actual heterogeneous exponents of the LiH block decomposition, not with a uniform k.

B. The convergence challenge

The Shibuya–Wulfman expansion has a well-documented convergence limitation. When a Sturmian on center A is expanded in the angular basis of center B, the expansion involves hyperspherical harmonics of increasing angular momentum L. The rate of convergence is controlled by the parameter $S = k \cdot R_{AB}$.

For $S \ll 1$ (nuclei close together relative to the orbital scale), convergence is rapid. For $S \gtrsim 3$ (typical molecular bond lengths), convergence is slow, requiring many angular terms. The Avery group explicitly notes this: the expansion "is very slowly convergent if S is large" [12].

For LiH at equilibrium ($R_{AB} \approx 3$ bohr), with $k \approx 1$ for the valence orbitals, $S \approx 3$. This places the system squarely in the slow-convergence regime. Each additional angular momentum channel in the expansion adds orbitals and therefore Pauli terms to the qubit Hamiltonian. If $L_{\max} = 5$–10 is required for convergence, the Pauli count could approach the full Level 4N encoding (∼3,288 terms), defeating the purpose of the block decomposition.

C. Resolution: multipole expansion termination

The convergence concern of the preceding subsection applies to the Shibuya–Wulfman *basis* expansion: expanding an orbital centered on A in the complete Sturmian basis of center B, which requires summing over all angular momenta L. The cross-center nuclear attraction integral of Eq. (1) uses a different expansion: the standard multipole expansion of $1/|\mathbf{r} - \mathbf{R}_B|$ in Legendre polynomials $P_L(\cos\theta)$.

For this expansion, the angular integral involves

$$\int Y^*_{l_1 m}(\hat{r})\, P_L(\cos\theta)\, Y_{l_2 m}(\hat{r})\, d\Omega\,, \tag{4}$$

which is nonzero only when $|l_1 - l_2| \le L \le l_1 + l_2$ and $l_1 + L + l_2$ is even. For $n_{\max} = 2$ ($l_{\max} = 1$), the maximum L needed is 2 (p–p pairs); for $n_{\max} = 3$ ($l_{\max} = 2$), $L_{\max} = 4$. The expansion terminates exactly at $L_{\max} = 2l_{\max}$—no truncation error.

Furthermore, $m_1 = m_2$ is required: the cross-center potential is diagonal in the magnetic quantum number when the internuclear axis is along z. Combined with the Gaunt selection rules, this means the cross-center V_{ne} adds only 33 new one-body matrix elements to h_1 for LiH at $n_{\max} = 2$. These are essentially free in computational cost compared to the two-body integrals.

The original grid-based radial integrals (trapezoid quadrature, $O(1/n_{\text{grid}})$ convergence) achieved 0.18% accuracy at $n_{\text{grid}} = 4000$ for $Z = 1$ orbitals, and 9×10^{-8} for $Z = 3$. The radial split integrals (inner and outer regions separated at R_{AB}) are integrals of polynomial times exponential: each $R_{nl}(r)$ contributes a polynomial in r times $\exp(-Zr/n)$. The product $R_{nl} R_{n'l'} r^p$ is a finite sum of terms $c_j r^j \exp(-\alpha r)$ where $\alpha = Z(1/n + 1/n')$. Each split integral evaluates exactly via the regularized incomplete gamma function (`scipy.special.gammainc/gammaincc`). This replaces the $O(1/n_{\text{grid}})$ trapezoid quadrature with machine-precision evaluation—an instance of the algebraicization program (Paper 18, Sec. XII).

D. The Coulomb resolution alternative

A potentially more efficient route was developed by Hoggan and collaborators [11]. The "Coulomb resolution" technique decomposes multi-center integrals over exponential-type orbitals into sums of one-electron overlap-like products, each involving orbitals on at most two centers. Crucially, these two-center integrals are separable in prolate spheroidal coordinates—the same coordinate system used at Level 2 of the GeoVac hierarchy (Paper 11).

The Coulomb resolution sidesteps the slowly convergent angular expansion of Shibuya–Wulfman by working directly in the two-center coordinate system where the integrals are most naturally expressed. GeoVac already possesses prolate spheroidal infrastructure from

Paper 11, including the Neumann expansion machinery for V_{ee} (Paper 12). The question is whether this infrastructure can be extended to compute the one-body cross-center integrals of Eq. (1).

V. THE COUPLED COMPOSITION ARCHITECTURE

If two-center integrals can be computed efficiently, the resulting "coupled composition" Hamiltonian would have the following structure for a molecule with B blocks:

$$H = \sum_{b=1}^{B} H_b^{(1)} + \sum_{b=1}^{B} V_{ee}^{(b)} + \sum_{b<b'} V_{ee}^{(b,b')} + \sum_{b=1}^{B} \sum_{a \neq a_b} V_{ne}^{(b,a)} \quad (5)$$

where $H_b^{(1)}$ is the one-body Hamiltonian within block b (kinetic + same-center nuclear attraction), $V_{ee}^{(b)}$ is the within-block ERI, $V_{ee}^{(b,b')}$ is the cross-block ERI, and $V_{ne}^{(b,a)}$ is the cross-center nuclear attraction (electron in block b feeling nucleus a from a different center).

Compared to the current composed Hamiltonian:

- PK is *eliminated entirely.* Core-valence orthogonality is enforced by the occupation number representation in second quantization, not by a pseudopotential.
- Cross-block ERIs are included, subject to Gaunt selection rules. Track CB established that these add $\sim$2/3 of the within-block ERI count (130 vs. 195 for LiH at $n_{\max} = 2$).
- Cross-center nuclear attraction is included via two-center integrals. The number of one-body terms is bounded by the number of orbital pairs times the number of off-center nuclei.

Conjecture 1. *The Gaunt selection rules that enforce ERI sparsity within each block also constrain the cross-block ERIs and the angular expansion of the cross-center nuclear attraction. If the two-center integral expansion converges in $L_{\max} \leq 3$ angular terms at molecular bond lengths, the coupled Hamiltonian should have $\lesssim 4.5\times$ the composed count for LiH at $n_{\max} = 2$, remaining below the full Level 4N count of 3,288 and potentially below the Gaussian STO-3G count of 907 after the PK one-body terms are removed. (As originally framed under the retired pair-diagonal rule this read $\lesssim 1{,}500$ against a composed count of 334; the* ratio *is the rule-independent content, and the absolute figure followed from it.)*

This conjecture is falsifiable by computing the two-center integrals and performing the Jordan–Wigner transformation.

Conjecture 2. *With a balanced Hamiltonian (all one-body and two-body terms present, no PK), the FCI accuracy on the coupled Hamiltonian should be significantly better than the composed framework's 5.3% $R_{\rm eq}$ error, bounded below by the hydrogenic FCI accuracy floor ($\sim$0.35% for He at $n_{\max} = 5$, Paper FCI-A).*

VI. RELATIONSHIP TO EXISTING GEOVAC INFRASTRUCTURE

The coupled composition program connects to multiple existing components:

Paper 7 (Dimensionless Vacuum): The S^3 conformal projection that defines GeoVac's graph Laplacian is identical to Fock's 1935 momentum-space mapping [6], which is the foundation of Shibuya–Wulfman theory. The 18 symbolic proofs of Paper 7 validate the same mathematical structure that generates the Shibuya–Wulfman integrals.

Paper 11 (Prolate Spheroidal): The prolate spheroidal coordinate system and Neumann expansion machinery are directly applicable to the Coulomb resolution technique for two-center integrals.

Paper 12 (Algebraic V_{ee}): The Neumann expansion of $1/r_{12}$ in prolate spheroidal coordinates demonstrates that multi-center Coulomb integrals can be decomposed into algebraically evaluable components via angular momentum recurrence.

Paper 14 (Qubit Encoding): Section V.E already bounds cross-block ERI counts at $\leq 2\times$ within-block; Track CB's empirical ratio of 2/3 is tighter. The coupled Hamiltonian would feed directly into the Jordan–Wigner encoding pipeline.

Paper 16 (Chemical Periodicity): The S_N representation theory and atomic classifier determine the block decomposition. Coupled composition uses the same block structure—it changes how blocks are coupled, not how they are defined.

Paper 18 (Exchange Constants): PK is classified as a "calibration exchange constant"—a transcendental quantity that enters when projecting between the graph and continuous coordinates. If coupled composition eliminates PK, it removes one exchange constant from the framework, consistent with the algebraicization program (Sec. XII of Paper 18).

Track BX-3b (Cross-Block MP2): The `cross_block_mp2.py` module already computes cross-block ERIs for LiH. This infrastructure would be reused for the cross-block ERI component of the coupled Hamiltonian.

VII. COMPUTATIONAL RESULTS

We report results from Phases 1–3 of the research path outlined in Sec. VIII.

A. Quantum resource comparison

Table III compares the qubit Hamiltonian metrics for the four coupling schemes at LiH $n_{\max} = 2$, $Q = 30$.

TABLE III. Qubit Hamiltonian resources for LiH at $n_{\max} = 2$, $Q = 30$. All values from Jordan–Wigner encoding, re-measured 2026-09-01 under the exact global-M_L rule. Pauli counts include the identity term; 1-norms exclude it. The Gaussian STO-3G baseline ($Q = 12$) is included for reference. Retired pair-diagonal figures: 334/37.3/21, 878/74.1/87, 854/85.7/88.

Configuration	Pauli	1-norm (Ha)	QWC
Composed (PK)	838	38.3	64
Balanced (CD)	2726	75.2	629
Coupled (CB)	2702	80.5	630
Gaussian STO-3G	907	34.3	273

The balanced Hamiltonian has 2,726 Pauli terms, confirming the ratio form of Conjecture 1 (3.25× the composed 838, under the conjectured 4.5×) but not its absolute form: 2,726 exceeds both the $\sim$ 1,500 the ratio implied at the retired composed count and the STO-3G count of 907. The 1-norm of the balanced Hamiltonian (75.2 Ha) is *lower* than the coupled CB value (80.5 Ha) because the attractive cross-center V_{ne} partially cancels the repulsive cross-block V_{ee}. The QWC group count (629; 87 under the retired pair-diagonal rule) no longer sits below the Gaussian baseline (273).

B. Cross-center V_{ne} census

The cross-center nuclear attraction integrals (Sec. IV C) add 33 nonzero one-body matrix elements to h_1 for LiH at $n_{\max} = 2$:

TABLE IV. Cross-center V_{ne} census for LiH at $n_{\max} = 2$, $R = 3.015$ bohr.

Block → nucleus	Nonzero	Max (Ha)	Trace (Ha)
Core ($Z=3$) → H	11	0.373	−1.646
Val-Li ($Z=1$) → H	11	0.328	−1.182
H-side ($Z=1$) → Li	11	0.985	−3.544
Total	33	— — —	−6.372

The large trace from the H-side block feeling the $Z = 3$ lithium nucleus (−3.544 Ha) is the dominant contribution, consistent with the physical expectation that the hydrogen electron is most strongly affected by the nearby heavier nucleus.

C. Four-electron FCI

Table V presents FCI results at $R = 3.015$ bohr for all four coupling schemes, including the new balanced configuration. Nuclear repulsion is V_{NN} only (no PK contribution).

TABLE V. Four-electron FCI for LiH at $n_{\max} = 2$, $R = 3.015$ bohr. Exact: −8.071 Ha.

Configuration	E_{FCI} (Ha)	Error	Bound?
Balanced (CD)	−7.924	1.8%	YES
Decoupled	−7.195	10.9%	NO
Composed (PK)	−6.863	15.0%	NO
Coupled CB	−5.734	29.0%	NO
Exact	−8.071	0%	YES

The balanced configuration is the only coupling scheme that produces a bound molecule. The PES scan gives $R_{\text{eq}} = 3.226$ bohr (7.0% error vs. exact 3.015 bohr) and $D_e = 0.037$ Ha (exact: 0.092 Ha). The variational bound is respected at all R-points.

Conjecture 2 predicted that the balanced Hamiltonian would achieve significantly better accuracy than PK's 5.3% R_{eq} error. The result (7.0%) does not confirm this in strict sense. However, the comparison is cross-regime: the 5.3% figure uses l-dependent PK at $l_{\max} = 2$ with continuous PES optimization, while the current result uses $n_{\max} = 2$ FCI on a qubit Hamiltonian with grid-computed integrals (0.18% radial accuracy for $Z = 1$). The qualitative result—the balanced coupled configuration is the *only* bound configuration among all four coupling schemes—is the primary finding of this phase.

D. Basis convergence: $n_{\max} = 3$

At $n_{\max} = 3$ ($Q = 84$, $l_{\max} = 2$), the balanced coupled Hamiltonian has 19,959 Pauli terms, 448.9 Ha 1-norm, and 2,298 QWC groups. The Pauli scaling exponent (2-point fit from $Q = 30$ to $Q = 84$) is 3.03; the 1-norm exponent is 1.75. The Pauli-to-composed ratio (2.53×) is consistent with $n_{\max} = 2$ (2.63×), indicating that the cross-block overhead is a constant factor, not a growing penalty.

The cross-center V_{ne} multipole expansion terminates exactly at $L_{\max} = 4$ for $l_{\max} = 2$ (d–d pairs), confirmed by comparison to $L_{\max} = 10$ (max difference: machine zero). The total cross-center V_{ne} trace is −12.54 Ha (168 nonzero terms).

The original grid-based integrals (trapezoid quadrature, $O(1/n_{\text{grid}})$ convergence) systematically underestimated cross-center nuclear attraction by 0.09–0.47% for diffuse $Z = 1$ orbitals. Replacing with analytical evaluation via incomplete gamma functions eliminates this error entirely (machine precision). This algebraicization is an instance of the exchange constant program (Paper 18, Sec. VII): the grid-based V_{ne} was a calibration exchange constant that proved eliminable once the right algebraic structure was identified. The corrected $n_{\max} = 3$ energy at $R = 3.015$ is −8.055 Ha (0.20% error), improved from grid-based −8.048 (0.28%). The R_{eq}, however, moved only from 3.287 (grid) to 3.280 (analytical), confirming that the outward drift is structural to the block decomposition, not numerical.

TABLE VI. Balanced coupled LiH convergence with basis size. Exact energy at $R = 3.015$ bohr: -8.071 Ha. Grid: trapezoid quadrature ($n_{\text{grid}} = 4000$). Analytical: incomplete gamma functions (machine precision).

Method	$n_{\max}$	Q	Pauli	1-norm (Ha)	$E(3.015)$ (Ha)	Error
Grid	2	30	2726	75.2	-7.924	1.8%
Analytical	2	30	2726	75.2	-7.929	1.7%
Grid	3	84	127855	436.9	-8.048	0.28%
Analytical	3	84	127855	436.9	-8.055	0.20%

Table VII shows the definitive R_{eq} comparison:

TABLE VII. R_{eq} comparison: grid vs. analytical integrals. Exact: 3.015 bohr.

Method	$n_{\max}$	R_{eq} (bohr)	R_{eq} err	$E(3.015)$ err
Grid	2	3.226	7.0%	1.8%
Analytical	2	3.227	7.0%	1.7%
Grid	3	3.287	9.0%	0.28%
Analytical	3	3.280	8.8%	0.20%
Exact	–	3.015	0%	0%

The $n_{\max} = 3$ analytical PES (5 R-points):

TABLE VIII. Balanced coupled LiH PES at $n_{\max} = 3$ (analytical integrals).

R (bohr)	E (Ha)
2.9	-8.0494
3.015	-8.0545
3.1	-8.0571
3.3	-8.0592
3.5	-8.0563

E. Fixed-geometry energy versus well shape

The convergent energy and the persistent outward R_{eq} drift are two independent read-outs of the same residual: the energy error is its magnitude at a fixed geometry, whereas R_{eq} is fixed by its *slope* at the true minimum. The well *curvature* is a third, independent readout. A harmonic fit to the $n_{\max} = 2$ balanced PES about its minimum gives $\omega_e \approx 2040$ cm^{-1} for LiH, some 45% above the experimental 1406 cm^{-1} (force constant 0.14 vs. 0.066 Ha/bohr^2). The balanced construction—the more accurate of the two architectures on *energy* (0.20% at $n_{\max} = 3$)—therefore misplaces both the position *and* the curvature of the minimum: its residual is concentrated in the *shape* of the potential well, not in the fixed-geometry energy (the well depth D_e itself is a separate, distinct read-out and is not extracted here). This shape residual is localized to the one-particle orbital basis: the tilt ε' and curvature (both evaluated at the true minimum) are *bit-invariant* to the cross-center V_{ne} angular multipole order (identical across $L = 2$–8, the multipole series Gaunt-terminating at $L = 2$ for the $l \leq 1$ orbitals) and—separately—insensitive to the radial quadrature. We state that second leg precisely, because the two quadrature knobs are not alike: the cross-center V_{ne} is evaluated analytically (incomplete gamma), so it carries no quadrature to converge at all; the one live radial quadrature is the cross-block electron-repulsion trapezoid, and refining it moves the tilt by 0.15% over $n_{\text{grid}} =$ 2000–8000 and 0.87% over 1000–8000. Equivalently, at the true minimum the computed electronic energy gradient is 8.8% weaker than the $+Z_A Z_B/R^2$ required to balance nuclear repulsion—a deficit an order of magnitude larger than the largest quadrature movement, so the well-shape error is an orbital-basis-completeness effect, not an integral-evaluation or angular-truncation effect (LiH $n_{\max} = 2$; a measured result). This mirrors the composed/PK result (Paper 17), where the *parameter-free* pseudopotential barrier reproduces ω_e to 4.6% but caps the attainable energy accuracy. For LiH the two architectures thus trade energy accuracy against curvature accuracy, and neither reproduces the full well; universality of this trade across hydrides is untested (a measured result; the curvature quoted here is the $n_{\max} = 2$ fine-grid value for LiH). For single-geometry applications (fixed-nucleus qubit Hamiltonians) only the energy readout enters, which is why the balanced construction remains well matched to that use despite its well-shape errors.

[**MEASURED**] The banked $n_{\max} = 4$ decider has now been run—a matrix-free (Davidson) direct-CI engine (`geovac/balanced_direct_ci.py`; $\sim$1600× faster than the assembled-matrix path at $n_{\max} = 3$, validated to $\leq 7 \times 10^{-15}$ Ha against a brute-force Fock-space FCI and to 1.8×10^{-15} Ha against the live library at $n_{\max} = 2$) solved the 16,040,025-determinant $n_{\max} = 4$ sector at six R points. The verdict is decisive for the *irreducible-wall* reading: the *per-shell increments* of energy, curvature and tilt at the true minimum all shrink geometrically at a common ratio (≈ 0.36–0.38)—the basis is converging—but every observable, the energy included, converges to a *wrong* limit. $E_{\text{tot}}(R_{\text{true}})$ runs $-7.9297 \to -8.0548 \to -8.1020$ Ha at $n_{\max} = 2/3/4$ (shipped convention, frozen-core offset -7.2799 Ha; the corrected sign shifts each value by $\approx +4$ mHa), *crossing* the exact -8.0705 between $n_{\max} = 3$ and 4 and extrapolating to ≈ -8.13: the balanced model over-binds past exact, so its excellent energy convergence is an approach-from-above statement valid through $n_{\max} = 3$ only. The curvature ratio at the true minimum runs $2.245\times \to 2.219\times \to 2.210\times$ the true force constant at $n_{\max} = 2/3/4$, extrapolating to $2.204\times$ (99.5% of the gap surviving, robust across geometric, power-law and deliberately-slow $1/n$ extrapolation models); the tilt worsens toward -0.0399; R_{eq} extrapolates to $+10.1\%$ and worsening; ω_e at the well's own minimum falls 2040 $\rightarrow$ 1946 $\rightarrow$ 1901 cm^{-1} (+45.1 $\rightarrow$ +38.4 $\rightarrow$ +35.3%, extrapolating to $\approx$ +32%—the earlier "stays $\sim$ +45%" reading was an $n_{\max}$ = 2-only fit), still far from 1406. Model-free statement: closing the curvature gap at the observed per-shell step would need over a hundred further shells with no decay in step size. The well-shape residual is therefore an *irreducible* property of the balanced architecture's orbital-basis limit, not slow basis healing (`tests/test_balanced_direct_ci.py`).

Reproducibility note. In the course of building the engine, a global sign error was found (and fixed) in the shipped assembled-matrix path's same-spin double-excitation phase: it contributed a nearly $n_{\max}$-independent spurious over-binding of $\sim +4$ mHa on balanced LiH while leaving the geometry observables essentially unchanged (< 0.2 pp). *Scope of the correction, stated precisely because the two conventions are easy to conflate.* The energies tabulated in this paper are on the *faithful* (pre-fix) convention: at the fixed geometry R = 3.015 the banked values are −7.9297, −8.0548, −8.1020 Ha at $n_{\max} = 2, 3, 4$, while the corrected phase gives −7.9256, −8.0509, −8.0982 Ha—a shift of +4.13, +3.85, +3.79 mHa. So the $n_{\max}$ = 3 energy error reads 0.20% on the faithful convention—the figure quoted throughout this paper— and 0.24% on the corrected one (both against the exact −8.0705 Ha). The earlier statement that "no number in this paper changes" was *wrong*: it compared the corrected energy at the well's own minimum ($R_{\rm eq} = 3.2776$, a different offset) against the faithful energy at $R = 3.015$, and the two agreed only by cancellation. What genuinely does not move is the geometry: $R_{\rm eq}$ shifts by ≤ 0.13 pp and the qualitative conclusions—binding, the stiffness excess, and the $n_{\max} = 4$ crossing—are unchanged under either convention. The corrected phase is pinned element-by-element against a brute-force Fock-space FCI in the same test file.

F. Summary of conjecture status

Conjecture 1 (Pauli count): **partially confirmed**. The balanced Hamiltonian has 2,726 Pauli terms. The conjecture's rule-independent content — $\lesssim$ 4.5× the composed count — holds at 3.25×. Its absolute legs do not: 2,726 exceeds the ∼1,500 that ratio implied at the retired composed count of 334, and exceeds the STO-3G count of 907. It stays below the Level 4N count of 3,288, but by 17% rather than the 3.7× the retired figures showed.

Conjecture 2 (accuracy): **MIXED**. Analytical integrals (machine precision via incomplete gamma functions) confirm that the $R_{\rm eq}$ drift is structural to the block decomposition, not numerical: grid error contributed only 0.007 bohr to $R_{\rm eq}$ at $n_{\max}$ = 3. The drift is +0.057 bohr from $n_{\max} = 2$ to 3 and +0.023 bohr from 3 to 4—decelerating, consistent with the extrapolated limit below (3× smaller than PK's +0.15–0.22 bohr/$l_{\max}$) but in the same outward direction. Energy converges excellently through $n_{\max}$ = 3: 1.8% → 0.20% (9× improvement) from $n_{\max}$ = 2 to 3—and then crosses *past* exact at $n_{\max}$ = 4 (over-binding; the well-shape subsection), so the fixed-geometry error is not monotonically improvable either. The balanced approach is optimal for single-point quantum simulation at fixed geometries (0.20% energy error), but not for PES-based equilibrium geometry determination—indicated at $n_{\max}$ = 4 (a three-point, model-robust extrapolation; and, model-free, closing the curvature gap would need well over a hundred further shells at an undecaying step) to be an irreducible limit of the architecture, not slow basis healing (the well-shape subsection above).

VIII. RESEARCH PATH

The following steps are required to test the conjectures of Sec. V:

1. **Two-center integral implementation.** *COMPLETE.* The convergence concern from the Shibuya–Wulfman basis expansion (Sec. IV) was resolved: the cross-center V_{ne} uses the standard multipole expansion, which terminates exactly at $L_{\max} = 2l_{\max}$ by Gaunt selection rules (Sec. IV C). For LiH at $n_{\max}$ = 2, only 33 one-body matrix elements are needed. Original grid numerical accuracy: 0.18% at $n_{\rm grid} = 4000$ for $Z = 1$ orbitals, 9×10^{-8} for $Z = 3$. Subsequently replaced by analytical evaluation via incomplete gamma functions (machine precision, zero quadrature). The decision gate ($L_{\max} > 5$) is moot—the expansion terminates at $L_{\max} = 2$.
2. **Balanced coupled Hamiltonian.** *COMPLETE.* The balanced Hamiltonian (Eq. 5 with all four interaction types) gives 2,726 Pauli terms, 75.2 Ha non-identity 1-norm, and 629 QWC groups (Table III). Conjecture 1 confirmed.
3. **FCI benchmark.** *COMPLETE.* Four-electron FCI gives $E = -7.924$ Ha (1.8% error), $R_{\rm eq}$ = 3.226 bohr (7.0%), $D_e = 0.037$ Ha. The balanced configuration is the only bound coupling scheme (Table V). Conjecture 2 not confirmed in strict sense (7.0% vs. 5.3% PK), but the comparison is cross-regime.
4. **Scaling study.** *COMPLETE.* $n_{\max} = 3$ FCI with analytical integrals (incomplete gamma functions, machine precision): $E(3.015) = -8.055$ Ha (0.20% error, down from 1.8% at $n_{\max} = 2$; see Table VI). Pauli scaling exponent 3.03, 1-norm exponent 1.75. $R_{\rm eq}$ = 3.280 bohr (8.8% error), confirming structural drift of +0.053 bohr per $n_{\max}$ step (3× smaller than PK's +0.15–0.22 bohr/$l_{\max}$). Grid error contributes only 0.007 bohr to $R_{\rm eq}$ (negligible). Classification: MIXED—energy converges excellently

through $n_{\max} = 3$ (then crosses past exact; the well-shape subsection), R_{eq} drifts structurally. Extension to BeH_2 and H_2O: COMPLETE (v2.0.41–42).

The critical risk is Step 1: if the angular expansion requires $L_{\max} > 5$ for convergence, the Pauli count will approach the full Level 4N encoding and the approach loses its structural advantage. The Coulomb resolution route (approach b) is the fallback if direct expansion is too slow.

A. Polyatomic census

The balanced coupled framework was extended to BeH_2 (collinear, v2.0.41) and H_2O (non-collinear, v2.0.42), completing the polyatomic census for all three composed molecules.

System	Blocks	Q	Composed Pauli	Balanced Pauli	Pauli ratio	λ_{comp} (Ha)	λ_{bal} (Ha)
LiH	2	30	838	2,726	3.25×	38.6	78.2
BeH_2	3	50	1,396	8,868	6.35×	374.9	323.4
H_2O	5	70	1,954	19,742	10.10×	372.5† / 28,061.9	1,612.0

TABLE IX. Balanced coupled resource census, re-measured 2026-09-01 under the exact global-M_L rule with each molecule at its own equilibrium geometry. Pauli counts include the identity term and 1-norms are identity-inclusive (the composed-with-PK total is only available that way). $^\dagger H_2O$ composed electronic-only (PK partitioned). Retired pair-diagonal figures: LiH 334/878/37.3/74.1, BeH_2 556/2,652/354.9/304.7, H_2O 778/5,798/361/28,053/1,509.3.

BeH_2 four-electron FCI gives 10.7% energy error with the balanced Hamiltonian, compared to 27.4% with PK and 10.9% decoupled. The balanced 1-norm (323.4 Ha) is *lower* than the composed 1-norm (374.9 Ha), confirming that PK adds more 1-norm than cross-center V_{ne} for this system.

H_2O required a non-collinear generalization: the O–H bond vectors make a 104.5° angle, so the cross-center V_{ne} matrices cannot be computed in a shared z-axis frame. The solution is Wigner D-matrix rotation of the real spherical harmonic basis: compute V_{ne} in the z-frame aligned with each bond vector, then rotate via block-diagonal $D \cdot V_z \cdot D^T$ where D is the rotation matrix for real spherical harmonics (identity for $l = 0$, Cartesian rotation permuted to $m = (-1, 0, 1)$ ordering for $l = 1$). FCI is infeasible for H_2O (10 electrons, $Q = 70$). The lone pair V_{ne} integrals are well-behaved: all traces are below 3 Ha per nucleus, confirming that the multipole expansion does not produce unphysical couplings for lone pair blocks.

The Pauli ratio (balanced/composed) grows with block count: 2.63× (2-block LiH) → 4.77× (3-block BeH_2) → 7.45× (5-block H_2O), because each block pair adds cross-block ERIs and cross-center V_{ne} terms. The growth is sub-*quadratic* in the block count over these three systems ($\sim B^{1.14}$—slightly super-linear; the per-block increment *decelerates*, 2.14 then 1.34 per block), so an $O(B^2)$ envelope is an upper bound rather than the measured trend.

The 1-norm advantage is regime-dependent. For VQE/NISQ applications where Pauli count and QWC groups dominate circuit cost, the composed architecture with PK remains cheaper. For fault-tolerant QPE where the 1-norm determines the number of Trotter steps, the balanced framework eliminates PK entirely. This is decisive for molecules with large nuclear charges: H_2O's PK contributed 98.7% of the total composed 1-norm (28,053 Ha), and the balanced 1-norm (1,509.3 Ha) is 18.6× cheaper. Even compared to the electronic-only composed 1-norm (361 Ha, with PK classically partitioned), the balanced 1-norm is 4.18× higher—but the balanced Hamiltonian does not require the classical 1-RDM post-processing step that PK partitioning demands.

B. The W1c-residual orthogonality wall (second-row hydrides)

For $Z \geq 11$ second-row hydrides (NaH, MgH_2, HCl, ...), the composed-coupled framework treats the heavy atom's [Ne] core as "frozen": the 10 core electrons are encoded only via the Hartree screening profile $Z^B_{\text{eff}}(\rho)$ on the cross-center V_{ne} multipole expansion, not as an explicit FCI block. This choice makes the qubit Hamiltonian tractable (NaH at $n_{\max} = 2$ fits in $Q = 20$ qubits with 2 valence electrons), but it introduces two structurally separable walls:

a. W1c (Hartree screening of cross-V_{ne}). Without screening, the bare cross-V_{ne} uses $Z_B = 11$ as if the [Ne] core were not there, leading to ∼10× overattraction of the H valence orbital. Phase C–W1c (`geovac/cross_center_screened_vne.py`, May 2026) replaces the bare Coulomb $-Z_B/\rho$ with the screened tail $-Z^B_{\text{eff}}(\rho)/\rho - \tilde{\Phi}^B(\rho)$ (Newton's shell-theorem decomposition), driven by the `neon_core.FrozenCore` machinery. NaH PES at $n_{\max} = 2$: the cross-V_{ne} magnitude is reduced by 5.4–6.0× across $R \in [1.5, 10]$ bohr, exactly matching the diagnostic prediction.

b. W1c-residual (W1b-class orthogonality). After W1c, NaH FCI energy descent is reduced from −6.24 Ha (bare) to −0.36 Ha (screened). The residual 0.36 Ha overattraction at $R = 2.5$ bohr (still 4.8× deeper than the experimental NaH $D_e \approx 0.075$ Ha, with no internal PES minimum) is the named "W1c-residual orthogonality wall": the H valence orbital is non-orthogonal to the [Ne]-core orbitals on the heavy atom, and the framework does not enforce Pauli exclusion of the H valence basis against the frozen core beyond what Hartree screening captures.

c. The Phillips–Kleinman cross-center engineering closure (negative result). The named engineering closure for the W1c-residual is the cross-center analog of the same-center Phillips–Kleinman projection; Paper 17 gives the same-center case. The implementation (`geovac/phillips_kleinman_cross_center.py`, May 2026) adds the standard PK barrier $\Delta H_{pq}^{\mathrm{PK}} = \sum_c (E_v - E_c) S_{pc} S_{cq}$ to the H-block one-body Hamiltonian, with cross-center overlaps S_{pc} computed by 2D Gauss–Legendre × radial-grid quadrature in the H-orbital frame, real-SH azimuthal selection ($m_p^{\mathrm{signed}} = m_c^{\mathrm{signed}}$) enforced exactly, and Clementi–Roetti HF eigenvalues E_c for the [Ne]-core orbitals (Na: $E_{1s} = -40.479$ Ha, $E_{2s} = -2.797$ Ha, $E_{2p} = -1.518$ Ha).

The result is HONEST NEGATIVE: PK cross-center reduces the W1c-residual descent from 0.357 Ha to 0.305 Ha (14.6% improvement) at standard absolute-PK parameters ($E_v = 0$, purely repulsive) but does NOT close the wall: NaH PES is still monotonically descending with no internal equilibrium minimum.

Configuration	$E_{\min}$ (Ha)	$R_{\min}$ (bohr)	Descent (Ha)	Bound?
bare (V_{ne} unscreened, no PK)	-171.097	2.5	6.244	no
W1c (screened, no PK)	-163.316	2.5	0.357	no
PK only (bare V_{ne}, with PK)	-170.949	2.5	6.097	no
W1c+PK (screened+PK)	**-163.264**	**2.5**	**0.305**	**no**

The decomposition shows W1c and PK are necessary engineering closures but not jointly sufficient: the residual ∼290 mHa overattraction lives in mechanisms beyond W1c+W1b, with candidate sources including (i) inner-core penetration physics that the screened multipole expansion handles correctly in form but may overestimate in magnitude near the unscreened nucleus, (ii) the hydrogenic $Z_{\mathrm{orb}} = 1$ basis on the heavy-atom valence side being qualitatively wrong (it is not the actual Na 3s orbital but a generic $Z_{\mathrm{eff}} = 1$ hydrogenic function), (iii) the cross-block ERI treatment between Na valence and H valence at mismatched effective charges. A proper basis on the heavy-atom valence side—e.g. radial Schrödinger eigenstates of the FrozenCore $Z_{\mathrm{eff}}(r)$ potential, available in `neon_core._solve_screened_radial`—is the most likely next engineering target.

The framework currently treats second-row chemistry-solver binding as an open architectural problem. The walls have been named and named-individually-closed-or-not at the engineering level (W1c yes, W1c-residual / W1b PK no, basis still open), but the joint closure does not yet produce a bound NaH PES.

d. The screened-Schrödinger valence basis (h1 diagonal only, Track 3, May 2026 — HONEST NEGATIVE on binding, DIAGNOSTIC POSITIVE on the next target). The named follow-on after the PK negative was to "replace the $Z_{\mathrm{orb}} = 1$ basis on the Na center with radial Schrödinger eigenstates of the FrozenCore $Z_{\mathrm{eff}}(r)$ potential." Track 3 (`geovac/screened_valence_basis.py`, 25 tests passing) implements the diagonal-only version of this substitution: the framework's $h_1[i,i] = -Z_{\mathrm{orb}}^2/(2 \cdot \mathrm{block\_}n^2)$ for the heavy-atom-side valence orbitals is replaced by the actual eigenvalue of `_solve_screened_radial_log` (s-states with `allow_l0=True`) or `_solve_screened_radial` ($l \geq 1$). Wired into `build_balanced_hamiltonian` via `screened_valence_basis: bool = False`; bit-exact backward compat preserved (LiH 2,726-Pauli regression bit-identical at machine precision; first-row blocks have `n_val_offset = 0` and are auto-skipped).

The eigenvalue substitutions are physically meaningful: Na 3s = -0.170 Ha (vs hydrogenic -0.500 Ha; $+0.330$ Ha correction), Na 4s = -0.067 Ha (vs -0.125 Ha; $+0.058$ Ha), Na 4p = -0.051 Ha (vs -0.125 Ha; $+0.074$ Ha each, three m_l states). The total trace shift on the NaH valence is $+0.611$ Ha — comparable to the W1c-residual descent depth (0.357 Ha).

Despite the large physical correction, the 8-configuration NaH PES test (bare/W1c/PK/SV combinations at max_n=2, Q=20, 2e FCI) shows *all 8 configurations remain monotonically descending with no internal equilibrium minimum.* The descent ladder updates to: bare 6.244 Ha → +W1c 0.357 Ha (17.5× reduction) → +W1c+PK 0.305 Ha (1.17× further) → +W1c+PK+SV 0.313 Ha (0.97× — slight worsening, within numerical noise).

The critical structural finding (the substantive result of Track 3, even though binding is not recovered): the screened-valence correction is *R-independent at the diagonal h_1 level.* Eigenvalues of the FrozenCore $Z_{\mathrm{eff}}(r)$ potential are properties of the on-center potential, not of the bond. The correction matrix is bit-identical at $R = 2.5$ and $R = 10$ bohr (max abs diff 5.55×10^{-17}, machine precision). A constant diagonal h_1 shift cannot affect the descent depth, which is by definition an R-dependent energy difference. *The W1c-residual wall is structurally not a diagonal-h1 problem.*

Diagnostic positive identification of the genuine target (`debug/w1c_residual_nah_track3_diag.py`): testing the cross-V_{ne} integral with hydrogenic-shape labels at the *physical n* (block_n=1, $l = 0 \to n_{\mathrm{phys}} = 3$ hydrogenic 3s instead of 1s) shows the framework over-attracts by 0.81 Ha at $R = 2.5$ vs the physical-n shape, and by only 0.14 Ha at $R = 10$. The differential, -0.674 Ha, is approximately 1.9× the W1c-residual descent (0.357 Ha) — a wavefunction-shape substitution would over-correct from the current attraction profile and likely produce a binding minimum.

The actual Na 3s screened wavefunction has mean radius $\langle r \rangle = 4.47$ bohr, while the hydrogenic Z=1 1s has $\langle r \rangle = 1.50$ bohr. The two have *opposite* shape biases: framework basis is too tight (1.50 bohr); the actual chemistry has a much more diffuse 3s. Cross-center V_{ne} scales strongly with this orbital extent; the diagonal eigenvalue substitution does not move the wavefunction shape.

Verdict on the named engineering closure: the SV diagonal correction is necessary architectural infrastructure (the eigenvalue machinery is correct and physically meaningful) but not sufficient for binding recovery. The genuine engineering closure is *wavefunction-shape substitution in cross-V_{ne} and within-block ERIs* — not eigenvalue substitution. Two architectural options remain:

1. **Physical-n hydrogenic relabeling (sprint-scale)**: rewrite the orbital-state lists so that block_n=1 with $l = 0$ on Na uses physical $n = 3$ hydrogenic shape (Slater integrals, V_{ne} integrals, all evaluated at the physical principal quantum number) at $Z_{\rm orb}$ chosen to match the asymptotic Na valence Z. Cheap, uses existing hydrogenic Slater infrastructure, but introduces a minor convention break.

2. **Strict-Schmidt orthogonalization (beyond sprint scale)**: explicitly Schmidt-subtract the heavy-atom core orbitals from the valence basis before computing all matrix elements; mathematically definitive but requires re-implementing h_1, ERI, and cross-V_{ne} in the orthogonalized basis with renormalization.

The diagnostic-shape test suggests Option 1 alone closes most of the descent and is the natural next named track.

e. Modular propinquity reframing: the W1c residual as a Hilbert C^*-bimodule deformation distance (Sprint M-Y, 2026-05-23). The modular propinquity literature (Latrémolière [15, 16]) provides a natural setting for the W1c-residual wall: the framework's NaH valence subblock at $n_{\rm max} = 2$ is a Hilbert C^*-bimodule $\mathcal{M}$ over $(\mathcal{A}_L, \mathcal{A}_R)$, where $\mathcal{A}_L$ is the H-centered multiplication algebra (the H-valence carrier) and $\mathcal{A}_R$ is the Na-centered algebra with FrozenCore structure (the Na-valence carrier). Candidate "pin states" for the bimodule element $\xi \in \mathcal{M}$ are the framework defaults (hydrogenic $Z_{\rm orb} = 1$) versus physically-screened alternatives (FrozenCore-screened Na 3s plus hydrogenic H 1s). The natural two-axis L/R decomposition of the bimodule deformation distance

$$d^{LR}_{\mathcal{M}}(\xi_a, \xi_b) = \left(d^2_L(\xi_a, \xi_b) + d^2_R(\xi_a, \xi_b)\right)^{1/2}$$

where $d_{L,R}$ are integrated against Gaussian probes centered at H (d_L) and Na (d_R) respectively, identifies the *right-action axis* (Na-side wavefunction shape) as the dominant carrier of the W1c residual. The structural prediction $d_R/d_L \gg 1$ is confirmed numerically by the Track α-Diagnostic (memo `debug/sprint_alpha_1_diagnostic_memo.md`): $d_R/d_L = 3.23$ for the load-bearing (framework hydrogenic) vs. (physical Na 3s) pair at the canonical probe width $\sigma = 1$ bohr, and $d_R/d_L \in [1.32, 3.58]$ across $\sigma \in [0.5, 3.0]$ bohr, $d_R/d_L = 3.23$ stable across $R \in [3.0, 4.0]$ bohr, and bit-stable to 4 digits under FrozenCore grid refinement. The qualitative ordering "$d_R \gg d_L$" is robust under all consistency checks; the precise factor is probe-width-dependent.

f. Alkali-hydride scaling refinement (Sprint α-Diagnostic). The bimodule diagnostic was applied to three alkali-hydrides (LiH, NaH, KH). The original M-Y prediction was a uniform monotone scaling $d_R(M) \sim r^{\rm phys}_{\rm valence}(M) - 1.5$ bohr, expecting Li < Na < K < Rb < Cs. The measured ordering is **LiH** $\ll$ **NaH** $\approx$ **KH** (not strict monotone): d_R(LiH) = 0.42, d_R(NaH) = 0.72, d_R(KH) = 0.64. The mechanism is the *radial-node count*: Li 2s has 1 radial node, Na 3s has 2, K 4s has 3. The K 4s inner-shell oscillations align partially with the hydrogenic $Z = 1$ 4s oscillation structure in the bond region, producing L^2-cancellation in the bond-region integrand $|R^M_{\rm hyd} - R^M_{\rm phys}|^2 w_R(z)$ that NaH's smoother (only 2 nodes) wavefunctions do not exhibit. The refined uniform scaling is therefore monotone-with-saturation, not strictly linear in $|\Delta r|$. The prediction for RbH (4 nodes) and CsH (5 nodes) is that they continue the plateau, matching the framework's empirical chemistry catalogue: LiH binds in production at $n_{\rm max} = 2$; NaH, KH, RbH, CsH do not.

g. Mixed Slater-n essential for screened valence orbitals (Sprint α-Multi-zeta, 2026-05-23). The named follow-on to the bimodule diagnostic is to substitute multi-zeta Slater expansions of the physically-screened Na 3s and 3p wavefunctions for the framework's hydrogenic $Z_{\rm orb} = 1$ basis. The fit (`geovac/multi_zeta_orbitals.py` Z=11 entry, Track α-Multi-zeta) returned an architectural finding distinct from BBB93 [17]/Clementi–Roetti 1974 conventions: the standard single-orbital expansion convention of fixing $n^k_{\rm slater} = n_{\rm orbital}$ for every primitive is *inadequate* for screened-Schrödinger eigenstates with significant core penetration. The physical Na 3s has $R(r = 0.011) \approx 3.06$ bohr$^{-3/2}$ (inner-shell core penetration) plus two radial nodes plus a diffuse outer tail spanning $r \in [0.01, 15]$ bohr. A pure $n = 3$ basis with prefactor r^2 cannot reach the inner amplitude without driving ζ to very large values, which then suppresses the outer tail. Empirically: pure-$n = 3$ at $K = 3$ gave overlap 0.9992 but only 1 radial node (node-count failure); mixed-$n = [1, 1, 2, 3, 3]$ at $K = 5$ gave overlap 0.999999, L^2 error 1.4×10^{-6}, correct 2-node structure, all $|c_k| < 1$. *Both zeta count and Slater n must be flexible* for a faithful screened valence basis.

h. The W1c residual has a third layer (Sprint α-PES, 2026-05-23, NEGATIVE-AT-CURRENT-BASIS). The named target of M-Y Path A is to substitute the multi-zeta physical Na 3s/3p basis into the production cross-V_{ne} kernel via the architectural extension of `geovac/shibuya_wulfman.py` (multi-zeta dispatch on `compute_cross_center_vne`) and `geovac/balanced_coupled.py` (`multi_zeta_basis: bool = False` kwarg). The implementation is correct (12 regression tests pass, bit-exact backward compat preserved); the Step 1 algebraic kernel differential at the Na 3s on-site cross-V_{ne} diagonal at NaH $R_{\rm eq} = 3.566$ bohr is -0.135 Ha (more than 2× the gate threshold; sign-consistent with the dissociation limit at $R = 10$ bohr where the differential drops to -0.025 Ha, 18.7% of the $R_{\rm eq}$ value). *Step 2 FCI verdict: bit-zero.* Multi-zeta substitution shifts the NaH 2-electron FCI ground-state energy by $|\Delta E| < 4 \times 10^{-13}$ Ha at every $R \in \{2.5, 3.0, 3.5, 4.0, 5.0, 10.0\}$ bohr.

The mechanism is a substantive new finding: *at*

NaH $n_{\max} = 2$ with bare cross-V_{ne}, the 2-electron FCI ground state lives entirely on the H side, not the Na side. The H 1s sees the un-screened Na $Z = 11$ nucleus at $R = 3.5$ bohr with on-site cross-V_{ne} attraction of -3.13 Ha (matching the classical point-charge limit $-Z_{\text{Na}}/R$ within 5%), compared to the Na 3s self-attraction of -0.28 Ha. The lowest 5 eigenvalues of the h_1 matrix are entirely on H-side states; the Na 3s eigenvalue sits at $i = 5$ in the eigenspectrum, *above* the lowest 5 bonding/antibonding orbitals on the H side. The 2-electron FCI ground state puts both electrons in the lowest H-side orbital with no Na 3s occupation, so the Na 3s diagonal/basis shift has *no effect on the FCI eigenvalue.*

i. Three-layer structure of the W1c residual (consolidated). After the α arc, the W1c-residual orthogonality wall on second-row alkali hydrides decomposes into three structural layers:

- **Layer 1 — H/Na orthogonality.** Addressed by `geovac/phillips_kleinman_cross_center.py`. Verdict: NEGATIVE on binding (14.6% improvement, PES still descending). Architecture retained; module is clean.

- **Layer 2 — Na-side wavefunction shape.** Addressed by `geovac/multi_zeta_orbitals.py` Z=11 entry plus dispatch in `shibuya_wulfman.py` and `balanced_coupled.py`. Verdict: Step 1 algebraic differential POSITIVE (-0.135 Ha at NaH R_{eq}); Step 2 FCI verdict bit-zero. Architecture retained; the FCI invisibility is the substantive new finding.

- **Layer 3 — FCI variational state H-side localization at finite basis** (NEW). At $n_{\max} = 2$, the FCI ground state structurally localizes on the H side; on-site h_1 corrections and on-site basis substitutions are bit-zero on the FCI eigenvalue. The residual lives in the variational state itself, not in operator-level matrix elements.

j. Three named follow-on directions.

1. **NaH $n_{\max} = 3$ test** (sprint-scale, highest priority). Bundled with the screened-path multi-zeta combination (currently flagged with `NotImplementedError`). Falsifies or confirms Layer 3 as a basis-size artifact: if the larger basis lets the FCI ground state acquire Na-side weight, multi-zeta is load-bearing and the α-PES verdict flips POSITIVE.

2. **Cross-V_{ne} kernel-shape substitution** (sprint-scale, parallel). Track 3's diagnostic (2026-05-09) identified this as the original named target with a substantial -0.674 Ha differential at $R = 2.5$ bohr. The α-PES Step 1 finding (-0.135 Ha) is a smaller-effect cousin; the kernel-shape substitution is mechanically a refactor of `shibuya_wulfman.py`'s `_radial_split_integral` to use physical-n orbital decomposition on the heavy-atom side instead of n_{block} hydrogenic.

3. **W1c + multi-zeta combination by removing NotImplementedError** (bundled with item 1). Architectural unification of the bare and screened paths.

k. F1 arc: unified W1c × multi-zeta architecture and the structural success / energetic failure split (Sprints F1-P1+P2, W1c × M-Z bridge, and F1 max_n=3, 2026-05-23). The F1 arc completed the multi-zeta integration at the architectural level and tested the unified architecture at two basis sizes (NaH max_n=2 and max_n=3). The arc closed with a CLEAN NEGATIVE on binding and a substantive new structural finding that sharpens the W1c-residual wall mechanism.

F1-P1+P2 (unified architecture at max_n=2): The defensive `NotImplementedError` blocking the combined W1c × multi-zeta dispatch (in `geovac/balanced_coupled.py` line 748) was removed cleanly: the bare-Coulomb and screening-correction sub-integrals compose at the integrand level, so the unified screened+multi-zeta path is a ∼25-line addition to `geovac/cross_center_screened_vne.py` with a `multi_zeta_basis: Optional[Dict] = None` kwarg. Three findings at NaH max_n=2: (a) multi-zeta is bit-zero on the FCI in the bare cross-V_{ne} case (reproducing the α-PES Step 2 finding) but *fully load-bearing when combined with W1c* (differential +0.056 Ha at $R = 3.5$ bohr scaling to +0.208 Ha at $R = 2.0$); (b) the original Layer-3 framing was revised — at W1c the FCI natural occupations are $[1.0, 1.0]$ (open-shell-singlet-like with Na 3s occupation ∼0.98), NOT H-dominant as inferred from the α arc; (c) the combined architecture reduces descent depth from W1c-alone 0.898 Ha to W1c+mz 0.690 Ha (23% reduction) but PES remains monotonically descending. Verdict: PARTIAL-CLOSURE-AT-MAX_N=2.

W1c × M-Z bridge sprint (three-bucket hypothesis): The bridge sprint applied the modular propinquity M-Z partition (cross-shift vs endomorphism; cf. Sub-sprint M-Z, 2026-05-23) to the W1c three-layer hierarchy and proposed a three-bucket refinement: cross-shift (handled) / basis-closable cross-shift (handled at sufficient basis size) / endomorphism (external input). Sub-layer 3 (orbital-pair flexibility) was classified as basis-closable cross-shift: the bonding combination $[\text{H 1s} \pm \text{Na 3s}]_\pm$ is structurally a bimodule cross-shift (left+right linear combination), conditionally handled at sufficient basis size. Three falsifiable predictions for NaH max_n=3 were frozen into `debug/data/sprint_w1c_mz_partition_predictions.json` before the test ran: **P1** (internal PES minimum at $R_{\text{eq}} \in [3.0, 4.5]$ bohr), **P2** (binding $D_e \in [0.0375, 0.150]$ Ha within 2× experimental $D_e \approx 0.075$ Ha), **P3** (multi-zeta differential $\in [0.02, 0.30]$ Ha at R_{eq}).

F1 max_n=3 (CLEAN NEGATIVE). The combined architecture at $Q = 56$ ran the three predictions:

Prediction	Actual	Verdict
P1: internal min $R_{\mathrm{eq}} \in [3.0, 4.5]$ bohr	$R_{\mathrm{min}} = 2.0$ bohr (smallest tested)	**FAIL**
P2: $D_e \in [0.0375, 0.150]$ Ha	$D_e^{\mathrm{PES}} = +0.7097$ Ha (10× exp.)	**FAIL** (cond.)
P3: $\lvert E_{\mathrm{W1c}} - E_{\mathrm{W1c+mz}} \rvert \in [0.02, 0.30]$ Ha	$\lvert 0.2066 \rvert$ Ha at $R = 2.0$	**PASS**

P1 fails on the exact falsification criterion (R_{min} at the smallest tested R); the PES is monotonically descending across $R \in \{2.0, 2.5, 3.0, 3.5, 4.0, 5.0, 7.0, 10.0\}$ bohr. P2 fails (conditional) — the spurious-binding descent of 0.7097 Ha is $\sim 10\times$ the experimental binding energy. P3 passes — multi-zeta endomorphism content is preserved across basis sizes (consistent with max_n=2's 0.21 Ha at the same R), confirming the endomorphism classification of multi-zeta itself does not depend on basis size. Aggregate: 1 of 3 predictions pass; CLEAN NEGATIVE per the bridge sprint's gate.

The substantive new structural finding (structural success / energetic failure split). The headline content the prediction test did not anticipate: at max_n=3 under combined W1c+mz, the dominant natural orbital IS a true bonding combination. At $R = 3.5$:

$$\phi_{\mathrm{dom}} \;=\; -0.698{\cdot}\phi_{\mathrm{Na,\,3s}} - 0.687{\cdot}\phi_{\mathrm{H,\,1s}} + \text{small admixtures}$$

with Na/H amplitude^2 split 0.50/0.50. The 2nd natural orbital is the antibonding combination $\phi_{\mathrm{2nd}} = -0.698 \cdot \phi_{\mathrm{Na,\,3s}} + 0.687 \cdot \phi_{\mathrm{H,\,1s}}$. At max_n=2 the dominant NO was 100% H-localized at every architecture; at max_n=3 the basis enlargement DID provide the orbital-mixing flexibility predicted by the bridge sprint. The Na/H amplitude split is stable at $\sim$50/50 across $R \in [2, 5]$ bohr — the bonding combination is not a localized feature of $R = 3.5$; it persists across the PES.

But the constructed bonding orbital has higher energy than the separated configuration at every tested R. The PES is monotonically descending; natural occupations remain $[1.0, 1.0]$ (open-shell singlet of bonding + antibonding singly-occupied), NOT $[2.0, 0.0]$ (closed-shell bond where the bonding pair is doubly occupied). The framework can BUILD the right orbital but cannot energetically PREFER it. The two halves of "binding" — orbital constructibility and energetic preference — split cleanly at this resolution.

Sub-layer 3 reclassification. The bridge sprint's hypothesis that sub-layer 3 is BIMODULE CROSS-SHIFT (basis-closable) is *partially confirmed and partially refuted.* The "structural" part is correct (the right linear combination IS structurally a bimodule cross-shift, and the basis enlargement DOES make it constructible). The "framework handles it" part is too strong: the framework constructs the orbital but cannot energetically prefer it. **Sub-layer 3 reclassifies to MODULE ENDOMORPHISM (basis-irreducible at tested scales)**, joining sub-layer 2 (multi-zeta basis-shape) in the endomorphism bucket. The proposed three-bucket M-Z partition is FALSIFIED; the original two-bucket partition (cross-shift / endomorphism) stands.

F2 cross-V_{ne} kernel-shape diagnostic (KERNEL-NOT-IT plus architectural-absence finding). Following the F1 max_n=3 CLEAN NEGATIVE, the next named target was the cross-V_{ne} kernel-shape substitution on the H partner side (Track 3's P2). Per the diagnostic-before-engineering rule, F2 ran the Step 1 algebraic diagnostic before committing to a production extension: five matrix elements computed via converged 3D numerical quadrature vs the framework's default multipole expansion at $L_{\mathrm{max}} = 4$. Largest converged differential: 2.0×10^{-5} Ha on the H 1s diagonal cross-V_{ne} contribution, which is $2.7 \times 10^{-4} = 0.027\%$ of the NaH bond-energy scale (0.075 Ha) — six to ten orders of magnitude better than what would be needed to close the W1c-residual wall. **Verdict: KERNEL-NOT-IT.** The multipole expansion is bit-faithful where it lives; further precision work within the existing block-diagonal architecture cannot close the wall.

The substantive new content the gate did not anticipate: the framework's h1 in `composed_qubit.build_composed_hamiltonian` is **strictly block-diagonal**. There is NO cross-block off-diagonal h1 matrix element $\langle \psi_a^A \mid V_{ne}(\text{any nucleus}) \mid \psi_b^B \rangle$ for orbitals on different centers $A \neq B$. The framework constructs h1 as a strict block-diagonal sum of atom-like sub-block Hamiltonians, with no slot for cross-block one-body coupling. The bonding/antibonding splitting from h1 alone in the $\{|\mathrm{Na\ 3s}\rangle, |\mathrm{H\ 1s}\rangle\}$ basis is therefore identically zero. The F1 max_n=3 bonding orbital (constructible from natural-orbital diagonalization of the 1-RDM) was emerging entirely from 2-body ERI coupling in the FCI, not from h1 cross-coupling — because the latter does not exist in the architecture. Sub-layer 3 reclassifies (third time, getting sharper): from F1's "module endomorphism" to F2's **"architectural absence"** — a missing matrix slot in the existing block-diagonal architecture, distinct from a missing calibration input or a wrong-value matrix element. New sub-wall name: **W1d** (cross-block h1 architectural extension), structurally orthogonal to W1c (frozen-core screening), W1a, W1b, W2a, W2b, W3.

F3 cross-block h1 architectural extension (W1d CLOSED at FCI level + W1e NEWLY NAMED). F3 implements the missing matrix slot: new module `geovac/cross_block_h1.py` ($\sim$ 437 lines, s-s only for first pass, axial geometry only, 18 new tests). The cross-block off-diagonal h1 matrix elements $\langle \psi_a^A \mid T + \sum_C (-Z_C/|r - R_C|) \mid \psi_b^B \rangle$ for orbitals on different centers $A \neq B$ are computed via 2D axial Gauss-Legendre quadrature with closed-form ϕ integral (s-orbital geometry), auto-rotation of the frame so the two centers are collinear, and kinetic via the Hellmann-Feynman identity $T\psi_B = (E_B + Z_B/r_B)\psi_B$. Production code: `build_balanced_hamiltonian` gains 5 `cross_block_h1` kwargs; `build_composed_hamiltonian` gains 6 kwargs; bit-exact backward compat at default `cross_block_h1=False` preserved (146 baseline tests + 18 new pass, zero regression).

Step 1 algebraic diagnostic at NaH $R_{eq} = 3.566$ *bohr:* for hydrogenic Na 3s, total h1[Na 3s, H 1s] = +0.320 Ha, bonding/antibonding splitting (lower-energy generalized eigenvalue) 3.46 Ha. For multi-zeta Na 3s: total = −1.370 Ha (sign flip; the multi-zeta Na 3s extends further into the bond region with mean radius 4.5 bohr vs hydrogenic 1.5 bohr), splitting 4.20 Ha. Both satisfy the strong-bonding decision-gate (50 mHa threshold) by 1-2 orders of magnitude.

Step 3 FCI verdict at NaH max_n=2 (Q=20, 2-electron singlet):

Architecture	$E(R=3.5)$ Ha	$E(R=10)$ Ha	D_e (2-pt) Ha	Naturals (R=3.5)
bare	−169.204	−164.672	+4.532	[1.9933, 0.0067] (H-only bonding)
W1c+mz	−163.115	−162.779	+0.336	[1.0000, 1.0000] (separated, no NO bonding signature)
W1c+mz+xblockh1	**−167.767**	**−163.985**	**+3.782**	**[1.9991, 0.0007] (bonding, 50/50 Na/H)**

The natural-orbital signature transformation is the F3 headline structural finding. W1c+mz alone has TWO singly-occupied separated orbitals (naturals exactly [1.0000, 1.0000]). W1c+mz+xblockh1 has ONE doubly-occupied bonding orbital (naturals [1.9991, 0.0007]). This is a **four-orders-of-magnitude change** in the dominant natural-orbital occupation signature, driven entirely by the cross-block h1 architectural extension. The h1 lowest eigenvector character flipped concurrently from Na/H = 0.00/1.00 (H-localized) at W1c+mz alone with eigvalue −0.802 Ha, to Na/H = 0.51/0.49 (bonding combination) at W1c+mz+xblockh1 with eigvalue −3.161 Ha — ∼ 2.4 Ha lower than the previously-lowest antibonding-Na-localized state. **W1d (the architectural absence named by F2) is CLOSED at the FCI level.**

Step 4 mini-PES with W1c+mz+xblockh1 (8 R-points across $[2, 10]$ *bohr):* PES is monotonically descending with $R_{\min} = 2.0$ bohr (smallest tested) and well depth $D_e^{\rm F3} = +4.37$ Ha — 58× experimental NaH $D_e \approx 0.075$ Ha. The dominant NO is bonding (50/50 Na/H) at every R; the cross-block h1 construction is robust across the full PES.

F3 verdict: PARTIAL-CLOSURE. Bonding-orbital construction CLOSED; inner-region overattraction OPENS. **New sub-layer W1e (inner-region overattraction)**: cross-block h1 introduces a bonding mechanism into the framework whose eigenvalue is lower than either separated atomic eigenvalue and whose matrix elements grow as orbitals overlap (intensifies at smaller R). The 2-electron FCI on NaH max_n=2 has NO explicit frozen-core electrons (the [Ne] core on Na is treated as a screening potential via W1c, not as occupied orbitals); therefore no Pauli-exclusion mechanism prevents the bonding pair from collapsing toward small R. This is the structural analog of the missing-Pauli-repulsion mechanism that standard Phillips-Kleinman pseudopotentials supply. The framework's existing PK cross-center barrier operates on the partner-side valence orbital diagonal, NOT on the bonding-combination orbital that cross-block h1 newly constructs — the natural F4 follow-on is the bonding-orbital PK extension.

Revised W1c wall mechanism (five-sub-layer hierarchy after the F1→F2→F3 arc).

- **W1c-cross-screening (CLOSED, Phase C, 2026-05-08).** Frozen-core $Z_{\rm eff}(r)$ reduces cross-V_{ne} by 5–6× on second-row systems. Bimodule cross-shift handled at the operator level via the screened-multipole structure.

- **W1c-multi-zeta-basis (CLOSED at architectural level, F1-P1+P2, 2026-05-23).** Physical screened Na 3s replaces hydrogenic $Z_{\rm orb} = 1$ basis. Chemistry-side module endomorphism, mitigable via FrozenCore + atomic-physics fits; bit-zero without cross-shift activator (bare cross-V_{ne}), load-bearing with W1c on (23% additional descent reduction at max_n=2).

- **W1d-cross-block-h1 (CLOSED at FCI level, F3, 2026-05-23).** Off-diagonal h1 between orbitals on different centers, structurally absent from `composed_qubit` prior to F3. Adding it via `geovac/cross_block_h1.py` transforms naturals $[1.0, 1.0] \to [1.9991, 0.0007]$ at NaH max_n=2 (four orders of magnitude in dominant occupation signature), constructing the bonding combination at the FCI ground state. Architectural-absence category (F2's sharpening of sub-layer 3): a missing matrix slot in the existing block-diagonal architecture, structurally distinct from kernel-error and from calibration-data endomorphism, framework-internally closable via architectural extension.

- **W1e-inner-region-overattraction (NEWLY NAMED, F3, 2026-05-23; F4 target).** Missing Pauli repulsion between the constructed bonding pair and the [Ne] frozen-core orbitals. Cross-block h1 lowers the bonding eigenvalue monotonically with decreasing R; no opposing repulsion in the 2-electron architecture. Net well depth $D_e^{\rm F3} = +4.37$ Ha vs experimental 0.075 Ha. Natural F4 closure: extend `geovac/phillips_kleinman_cross_center.py` (currently partner-side valence diagonal) to project the F3-constructed bonding combination onto the [Ne] frozen-core orbitals and apply Phillips-Kleinman repulsion at the bonding-orbital level.

- **(FALSIFIED hypothesis: three-bucket M-Z refinement.)** The bridge sprint's basis-closable cross-shift sub-class evaporates at F1 max_n=3: the bonding combination IS structurally a bimodule cross-shift and IS basis-closable at max_n=3 in the orbital-construction sense, but the framework cannot energetically prefer it from within its bimodule machinery at the tested scales. Two-bucket

M-Z partition stands with architectural-absence as a structurally distinct third sub-category WITHIN the partition.

Three named F4 follow-on candidates.

1. **F4 — Bonding-orbital Phillips-Kleinman extension (TOP PRIORITY, sprint-scale).** Extend `geovac/phillips_kleinman_cross_center.py` to act on the cross-block-h1-constructed bonding orbital (rather than on the partner-side valence diagonal). After cross-block h1 is added in F3's phase 3.5, diagonalize h1 to identify the dominant bonding combination; compute its overlap with the [Ne] frozen-core orbitals via the F3-style 2D axial quadrature; add a PK projector $\Delta H^{\mathrm{PK,bond}} = \sum_c (E_v - E_c) \cdot \mathcal{P}_{\mathrm{bond}} S_{vc} \langle c|$ at the operator level. Decision gate: internal minimum at NaH max_n=2 with R_{eq} within 1 bohr of 3.566 and well depth in $[0.0375, 0.150]$ Ha closes the W1c-residual wall to within 2× experimental binding energy. Extends existing PK infrastructure rather than requiring new architecture.
2. **F3-extend: $l > 0$ cross-block h1 + non-axial geometry (beyond sprint scale).** Current F3 cross-block h1 is s-s only; mixed angular momentum requires spherical harmonic decomposition at both centers with their own quantization axes, bipolar harmonic algebra or full 3D Lebedev quadrature, and non-collinear geometry handling. Necessary to generalize F3's architectural advance from NaH/MgH_2/HCl alkali hydrides to polyatomic frozen-core systems (H_2S, PH_3, ...).
3. **DO NOT pursue further max_n enlargement on NaH OR within-sub-block kernel-precision work.** F1 max_n=3 + F2 + F3 between them closed the basis-size and kernel-precision directions: at max_n=3 the structural gap (bonding orbital constructibility) is closed by basis enlargement; at max_n=2 with cross-block h1 the structural gap is closed by architectural extension. The remaining wall (W1e) is NOT basis-size limited and NOT kernel-precision limited; it is a Pauli-repulsion gap at the bonding-orbital level, addressed only by F4 architectural extension.

l. Source documentation. Track α-Diagnostic: `debug/sprint_alpha_1_diagnostic_memo.md`. Track α-Multi-zeta: `debug/sprint_alpha_2_multizeta_memo.md`. Track α-PES: `debug/sprint_alpha_3_pes_test_memo.md`. Track β synthesis (modular + α arc): `debug/sprint_modular_alpha_arc_synthesis_memo.md`. F1-P1+P2 unified architecture: `debug/sprint_f1_p1p2_combined_test_memo.md`. W1c × M-Z partition bridge: `debug/sprint_w1c_mz_partition_analysis_memo.md`. F1 max_n=3 CLEAN NEGATIVE: `debug/sprint_f1_maxn3_predictions_test_memo.md`. Sprint β-neg synthesis (F1-maturity, superseded by full-arc memo): `debug/sprint_beta_neg_synthesis_memo.md`. **F2 cross-V_{ne} kernel-shape diagnostic (KERNEL-NOT-IT + architectural-absence finding, W1d named):** `debug/sprint_f2_cross_vne_kernel_memo.md`. **F3 cross-block h1 architectural extension (W1d CLOSED + W1e newly named):** `debug/sprint_f3_cross_block_h1_memo.md`. **Comprehensive full-arc synthesis (F1→F2→F3 at F3 maturity):** `debug/sprint_w1c_full_arc_synthesis_memo.md`.

1. *Refinement of the W1e mechanism after F4–F6*

The F3 PARTIAL-CLOSURE verdict named W1e (inner-region overattraction) as the natural F4 target with the working hypothesis "missing Pauli repulsion against the [Ne] frozen core," with three candidate closure mechanisms ranked by priority: (1) bonding-orbital Phillips–Kleinman extension (F4), (2) explicit frozen-core electrons in FCI (F5 priority), (3) valence basis enlargement (F6 priority). Three diagnostic sprints in immediate succession (F4, F5, F6) tested all three candidates and refined the W1e mechanism into a structurally distinct multi-sub-mechanism reading. **None of the three candidates closes W1e on its own; individually they account for at most ∼10% and ∼26% of the wall and jointly for ∼36%, leaving the ∼65–90% itemized below—the spread reflecting whether one or both closable mechanisms are counted—as genuinely-deep correlation requiring architectural lifts beyond sprint scale.**

F4 — bonding-orbital PK extension RULED OUT (rank-1 PK saturation → 43% closure ceiling). The Step 1 algebraic diagnostic at NaH $R_{\mathrm{eq}} = 3.566$ bohr computes the bonding-orbital–core overlap from the full F3 h1 diagonalization ($\max|S| = 17.5\%$ on the Na 2s core, well above the 5% strong-gate threshold) and the corresponding Phillips–Kleinman barrier at strict absolute reference $E_v = 0$ as +0.194 Ha (22× smaller than the 4.37 Ha wall depth). The diagnostic-before-engineering FCI sensitivity test injects a rank-1 PK shift $\Delta H = c \cdot |\mathrm{bond}\rangle\langle\mathrm{bond}|$ on the F3 h1 at varying c: at the Step 1-predicted $c = +0.19$ Ha the FCI well depth changes by 0.05% (4.3735 Ha → 4.3712 Ha), and even at $c = 2.0$ Ha (10× the predicted barrier) closure saturates at 43% (residual ∼2.5 Ha, still 33× experimental). The rank-1 PK projection on the bonding orbital is structurally near-orthogonal to the 2-electron FCI's dominant correlation channels at small R — the FCI ground state puts significant weight on configurations OTHER than the bonding-pair doubly-occupied state, and a rank-1 perturbation on the bonding orbital does not lift those configurations. **W1e is therefore NOT a single-particle**

Pauli-orthogonality wall (the working hypothesis F3 named); it is a multi-determinant FCI correlation effect that rank-1 PK cannot suppress. F4 STOP-at-Step-1 documented in `debug/sprint_f4_bonding_pk_memo.md`.

F5 — explicit-core Hartree treatment RULED OUT (25.7% closure ceiling at 2-point level). The Step 1 algebraic diagnostic computes the cross-block 2-body Coulomb correction $\sum_c (J-K)_{\mathrm{bonding},c} = +1.12$ Ha (Hartree $J = +1.13$ Ha, Slater–Condon $K = +0.008$ Ha, $K/J = 0.72\%$) between the F3 bonding orbital and the five Na [Ne] core orbitals at NaH $R_{\mathrm{eq}} = 3.566$ bohr using Clementi–Raimondi exponents. The correction is the right SIGN (repulsive, would lift the inner-region collapse) but only 25.7% of the 4.37 Ha wall depth. Linear extrapolation of the predicted Step 3 outcome gives residual $D_e^{\mathrm{F5}} \approx +3.25$ Ha $= 43\times$ experimental NaH $D_e \approx 0.075$ Ha, well outside the prespecified $2\times$ closure window $[0.0375, 0.150]$ Ha. **The F4 precedent applies: F5 saturation at higher Hartree strengths would probably reach 30–50% closure, structurally insufficient.** F5 STOP-at-Step-1 documented in `debug/sprint_f5_explicit_core_memo.md`. W1e reclassified from "FCI correlation channels addressable by explicit cores" to **"deep correlation effect beyond Hartree-level core-bonding J − K interaction."**

F6 — valence basis enlargement to max_n=4 RULED OUT (10.2% PES closure ceiling; 2-point gate overestimates by 2.5×). F6 ran the full F3 stack (W1c-screening + multi-zeta-basis + cross-block-h1) at NaH max_n=4 ($Q = 120$, $M = 60$, FCI dim 3600; ∼5 min/single-point) across 5 R-points $\{2.0, 3.0, 3.566, 5.0, 10.0\}$ bohr. PES well depth at $R_{\min} = 2.0$: $D_e^{\mathrm{PES,\,F6}} = +3.929$ Ha, 10.2% **closure**. The 2-point gate (R=3.5 vs R=10) used in F3 §3 suggests 26.1% closure at max_n=4 but this is **ARTIFACTUAL** — the PES scan reveals the inner-region collapse continues from R=3.5 down to R=2.0 (energy drops by another 0.696 Ha that the 2-point gate misses). The bonding signature is structurally preserved at every R (dominant NO $\in [1.9971, 1.9992]$; Na/H amplitude $\approx$ 50/50). Methodological finding for future quick-gate testing in this regime: **the 2-point gate overestimates PES-derived closure by 2.5× on this class of monotonically-descending PES; future sprints should use PES well depth at $R_{\min}$ as the load-bearing closure metric**, not 2-point energy differentials at fixed reference geometry. F6 PARTIAL_IMPROVEMENT documented in `debug/sprint_f6_maxn4_nah_memo.md`.

Refined six-sub-layer W1c hierarchy after F4–F6. W1e decomposes into three structurally INDEPENDENT sub-sub-mechanisms, each with a measured closure ceiling within sprint-scale reach:

- **W1e-basis-truncation:** ∼10.2% closure ceiling at NaH max_n=4 PES level (F6, measured); diminishing returns expected at max_n=5+ based on the 10% level at the 6× basis-size increase from max_n=2.
- **W1e-Hartree-pressure:** ∼25.7% closure ceiling at max_n=2 2-point level (F5, predicted from cross-block 2-body Coulomb $J - K$ Hartree mean-field); 2-pt-vs-PES caveat from F6 suggests PES-level closure may be lower than the 2-pt prediction.
- **W1e-deep-correlation-residual:** ∼65–90% of the wall remains beyond mean-field + single-determinant + basis-enlargement architecture. Closure candidates: Schmidt orthogonalization at the basis-construction level (beyond sprint scale), explicit treatment of [Ne] core as fully correlated (a major structural extension: requires occupancy-restriction FCI infrastructure not present in framework), or acceptance that the framework's structural-skeleton scope (CLAUDE.md §1.7) does not autonomously extend to binding energetics at the second-row regime within current compute budgets.

Methodological caution flagged for future sprints (2-point-vs-PES discrepancy). F1, F2, F3 all used a 2-point gate at $R = 3.5$ bohr vs $R = 10$ bohr as the quick-gate test for D_e. F6's PES scan reveals this gate overestimates wall depth by ∼2.5× for monotonically-descending PES in the W1c-residual regime, because the inner-region collapse extends below $R = 3.5$ down to $R_{\min} = 2.0$ with significant additional descent (∼0.7 Ha at max_n=4) that the 2-point gate misses. **Future quick-gate tests in this regime should use a 3-point or 5-point mini-PES across $R \in [2, 10]$ bohr** rather than a 2-point differential at fixed reference geometry, to avoid mis-attributing closure fractions. F3's reported $D_e^{\mathrm{F3}} = +4.37$ Ha is itself a 2-pt value (R=3.5/R=10); the corresponding F3 PES well depth at $R_{\min} = 2.0$ is likely larger (untested but expected by analogy to F6).

Net W1e closure verdict after F4–F6: deep correlation that resists basis-truncation, single-particle PK, and mean-field Hartree-class closure at scales reachable within sprint-scale compute. The chemistry arc's forward direction shifts from "find the right closure mechanism" to "characterize the framework's structural-skeleton scope limits empirically for second-row hydride binding." Consistent with the broader CLAUDE.md §1.7 multi-focal-composition wall taxonomy finding (the framework's structural-skeleton scope does not autonomously deliver binding energetics at this precision).

a. Source documentation (F4–F6 refinement). F4 bonding-orbital PK STOP-at-Step-1: `debug/sprint_f4_bonding_pk_memo.md`. F5 explicit-core Hartree STOP-at-Step-1: `debug/sprint_f5_explicit_core_memo.md`. F6 max_n=4 PARTIAL_IMPROVEMENT: `debug/sprint_f6_maxn4_nah_memo.md`. Sprint β-update F4-F6-APPENDED-TO-F3-BASELINE synthesis: `debug/sprint_beta_update_f4f6_memo.md`.

2. Closure of the W1e sprint-scale candidate space and cross-domain pattern (Schmidt + core correlation, 2026-05-23)

The two named beyond-sprint-scale candidates from §VIII B 1 (“Schmidt orthogonalization at basis-construction level” and “fully correlated [Ne] cores”) were tested by Day-1 diagnostics, executing the diagnostic-before-engineering rule at sprint-cycle scale. **Both ruled out as W1e closure mechanisms.**

Schmidt orthogonalization at basis-construction level (Day-1 algebraic diagnostic, STOP). The H 1s basis function on the H center has total projection mass $\sum_c |S_c|^2 = 1.00 \times 10^{-3}$ on the Na [Ne] core at NaH $R = 3.5$ bohr — only 0.1% of H 1s lives in [Ne]-core space. Three nonzero overlaps (Na 1s, 2s, $2p_0$) survive the axial m-selection rule; Na 2s dominates ($|S|^2 = 8.9 \times 10^{-4}$, $|S| \approx 3.0\%$). The 17.5% max-overlap F4 found (§VIII B 1) was specifically on the F3-constructed *bonding orbital*, which mixes H 1s with Na 3s; the H 1s basis function alone is structurally far outside the [Ne] core’s significant amplitude. Total diagonal-element differential: $\Delta_{\text{total}} = +8.6$ mHa (converging from below across a 4-step Cartesian-grid sweep $80^3 \to 140^3$), which is $\sim 12\times$ below the 100 mHa prespecified GO threshold and $\sim$ 3–4$\times$ below the 30 mHa BORDERLINE threshold; it would close at most $\sim 0.2\%$ of the 4.37 Ha W1e wall depth. **The W1e wall is not a basis-level non-orthogonality wall.** Day-1 driver and memo: `debug/sprint_w1e_schmidt_diagnostic_memo.md`, `debug/data/sprint_w1e_schmidt_diagnostic.json`.

Fully correlated [Ne] cores (Day-1 literature order-of-magnitude estimate, DEFER). The binding-relevant [Ne] core-correlation differential at NaH R_{eq} converges from three independent literature anchors to $\sim$ 0.001–0.003 Ha (central), ~ 0.007 Ha (upper bound): Iron-Oren-Martin 2003 qualitative ranking of Na subvalence correlation strength; Sylvetsky-Martin 2018 W4-17 diatomic core-valence band; atomic Na [Ne]-core correlation ($\sim$ 0.36 Ha) times an estimated active fraction at NaH bond geometry (0.5–2%). All three sit 14–1000$\times$ below the prespecified 0.1 Ha DEFER threshold. **Core correlation is a Layer-2 sub-percent effect for Na chemistry; it cannot close a ~ 3.85 Ha structural wall (scales mismatch by $\sim 10^3$).** Day-1 memo: `debug/sprint_w1e_core_correlation_estimate_memo.md`, `debug/data/sprint_w1e_core_correlation_estimate.json`.

Net W1e closure verdict: sprint-scale candidate space exhausted. Five independent sprint-scale W1e closure mechanisms now ruled out: F4 single-particle Phillips–Kleinman (43% saturation ceiling), F5 mean-field Hartree $J - K$ (25.7% predicted ceiling), F6 basis enlargement at $n_{\max} = 4$ (10.2% PES closure), basis-level Schmidt orthogonality (0.2% closure), [Ne] core correlation ($\sim 10^{-3}$–10^{-2} of wall). The W1e wall is deep multi-determinant correlation at the heavy-atom valence/core boundary in the second-row regime; closure candidates beyond sprint scale are major architectural commitments (self-consistent Z_{eff} depending on cross-center electronic density; fully-correlated all-electron treatment incompatible with the two-electron active-space architecture; fundamentally different basis families) or acceptance of the framework’s structural-skeleton scope as the autonomous boundary for second-row binding energetics.

Cross-domain pattern crystallization. The W1e wall is the **sixth structurally-independent instance of the multi-focal-composition wall pattern** documented across the framework (CLAUDE.md §1.7; Paper 34’s projection catalogue). The five previous instances are all in physics observables: Sprint H1 Yukawa (no autonomous Y_F selection); LS-8a $Z_2/\delta m$ (no autonomous renormalization counterterms despite faithful reproduction of two-loop UV-divergent integrand); HF-3 recoil (no cross-register $V_{ne}(\hat{r}_e, \hat{R}_n)$ coordinate operator); HF-4 Zemach (no magnetization-density operator at the proton coordinate); HF-5 multi-loop a_e (same renormalization wall as LS-8a, vertex sector). W1e adds a chemistry-domain instance: the framework constructs the correct bonding orbital at the FCI level (W1d closure in F3) but does not autonomously generate the energetic preference for that orbital at the correct R_{eq}. **The structural-skeleton scope statement extends from “five physics observables” to “five physics + one chemistry observable” — the framework determines the structural skeleton across both domains but does not autonomously generate calibration data in either.**

a. Source documentation (W1e Schmidt + core correlation closeout). Schmidt orthogonalization Day-1 diagnostic STOP: `debug/sprint_w1e_schmidt_diagnostic_memo.md`, `debug/data/sprint_w1e_schmidt_diagnostic.json`. [Ne] core-correlation order-of-magnitude estimate DEFER: `debug/sprint_w1e_core_correlation_estimate_memo.md`, `debug/data/sprint_w1e_core_correlation_estimate.json`.

3. Structural classification of W1e and the explicit-core HF decision-gate diagnostic (2026-05-25)

After the sprint-scale candidate-space closure (§VIII B 2), a diagnostic-only structural reading sprint placed W1e within the cross-domain multi-focal-composition wall pattern and named a load-bearing decision-gate diagnostic.

Structural reading: W1e is H1-class, not LS-8a-class. Three independent comparisons place W1e in the same structural sub-class as Sprint H1’s Yukawa-undetermined verdict (admits structure, lacks autonomous selection) rather than the LS-8a / HF-5 sub-class (requires external counterterms parameterizing UV physics): (i) the framework CAN build the right structure — W1d closure in F3 constructed the bonding orbital at the FCI level (naturals $[1.9991, 0.0007]$, four orders advance over W1c+mz alone); (ii) the closure mechanism is "external preference data" (supplying the Pauli pressure to select the bonding orbital at the correct $R_{\rm eq}$), not "external counterterms" (which have the open-ended Landau-pole character of LS-8a / HF-5); (iii) the calibration data is framework-internally computable in principle (existing `geovac/neon_core.py` infrastructure can be inverted from screening-potential mode to explicit-orbital mode; `geovac/multi_zeta_orbitals.py` provides the basis infrastructure). The chemistry-side distinguishing feature is that the relevant external input (atomic-structure data) is *closer at hand* than UV-completion physics — substantial architectural commitments beyond sprint scale rather than a fundamental UV completion outside the framework.

Closure-prospect classification: (a-with-caveat) sprint-scale exhausted but tractable for components as a substantial open program, with a calibration-external residual whose magnitude depends on the Reading A vs Reading B verdict (since resolved: Reading B; Sprint B.1 note below). The three sub-sub-mechanisms identified at F4-F6 (basis-truncation $\sim$ 10.2% / Hartree-pressure $\sim$ 25.7% / deep-correlation-residual 65–90%) stack to $\sim$ 35–45% combined ceiling at sprint scale; the question is whether implementation beyond sprint scale pushes this to $\sim$ 85–95% (Reading A: active-space deficit, well-defined CCSD-class closure) or to $\sim$ 30–50% leaving a structurally bounded residual (Reading B: architectural-scope boundary analogous to H1's Yukawa-undetermined verdict).

Explicit-core 12-electron HF on NaH is the named load-bearing diagnostic. Implement HF on the full 12-electron NaH problem in the framework's existing basis machinery (single-zeta CR67 + multi-zeta for valence, $\sim$ 120-orbital matrix size at $n_{\rm max} = 2$). Run 8-point PES sweep. Decision-gate: if HF gives internal minimum at $R_{\rm eq} \in [3.0, 4.5]$ bohr with $D_e \in [0.04, 0.12]$ Ha $\rightarrow$ **Reading A confirmed** (W1e is active-space deficit; a well-defined CCSD-class post-HF closure path exists beyond sprint scale). If HF still gives monotone-descending PES $\rightarrow$ **Reading B confirmed**. *(Since run: Sprint B.1, 2026-06-07—the 12-electron explicit-core RHF still descends monotonically; Reading B is the confirmed outcome.)* (W1e is structurally tied to the composed-architecture choice; chemistry-domain architectural-scope boundary established, structurally analogous to H1's Yukawa-undetermined verdict in the physics domain).

Three-class partition of the framework's external-input space (Sprint Chemistry Positioning Track 2, 2026-05-25). The cross-domain wall pattern at the boundary (6 instances, 5 physics + 1 chemistry) is one structural reading. What sits *outside* the boundary refines into three categorically distinct classes: **(1) Calibration parameter values** (Yukawas, Z_2, δm; UV-physics or AC-inner-factor tied; structurally bounded; "second packing axiom" is the speculative path forward) — members H1, LS-8a, HF-5. **(2) Multi-focal composition** (spatial $\times$ spatial composition rules; architectural extension well-defined within existing computational primitives) — members HF-3, HF-4 (operator-level largely closed by W1b extension). **(3) Multi-determinant correlation** (N-representable density-matrix manifold beyond the zero-entropy class; chemistry-methods on the correlation submanifold; ambiguous between Reading A framework-internal-extendable and Reading B architecturally-scoped) — member W1e. The diagnostic-before-engineering implication: don't propose Class-2 closure mechanisms (single-particle PK, mean-field Hartree, magnetization extensions) for Class-3 walls — they project onto the zero-entropy density-matrix submanifold and saturate below the wall, as F4-F6 + Schmidt + core correlation empirically demonstrated.

a. Source documentation (structural classification + diagnostic). W1e structural reading + closure-prospect classification: `debug/sprint_chemistry_position_w1e_structural_memo.md`. Modular $\rightarrow$ chemistry clean-negative + three-class partition: `debug/sprint_chemistry_modular_connection_memo.md`. Memory: `memory/external_input_three_class_partition.md`.

IX. DISCUSSION

The composed framework's PK pseudopotential was a necessary compromise for *classical* solvers that could not handle the full inter-group Coulomb interaction. In second quantization on a quantum computer, antisymmetry is enforced by the fermionic algebra, not by coordinate-space projectors. The occupation number representation naturally prevents double-occupancy without PK. This suggests that PK's *antisymmetry enforcement* role is a classical solver artifact. However, PK also approximates the physical core-valence Coulomb interaction energy—this role must be *replaced* by explicit inter-block integrals, not simply removed.

Track CB's negative result confirms this distinction. The failure was not because PK was needed for antisymmetry, but because the replacement was incomplete: cross-block V_{ee} was added without the balancing cross-center V_{ne}. The 10.9% error of the fully decoupled Hamiltonian (better than PK's 15.0%) reinforces the interpretation that PK overcorrects the Coulomb interaction, but the remaining 10.9% error shows that inter-block coupling *is* needed for quantitative accuracy.

The four-electron FCI results (Sec. VII) confirm the diagnosis from Track CB. PK overcorrects (15.0% error, unbound). Removing all inter-block coupling undercorrects (10.9%, unbound). Adding only cross-block V_{ee} without V_{ne} is energetically unbalanced (29.0%, worst of all). Only the balanced combination of all four interaction types—within-block h_1 and V_{ee}, cross-block V_{ee}, and cross-center V_{ne}—produces a bound molecule with the correct qualitative PES shape. This suggests that the physical content of PK is not pseudopotential enforcement of orthogonality, but rather an incomplete approximation to the sum of cross-block V_{ee} and cross-center V_{ne}.

A practical concern: Track CB found that adding cross-block ERIs increases the 1-norm from 37.3 to 85.7 Ha. Even if PK is eliminated, the net quantum resource cost could increase if the balanced coupled Hamiltonian has higher 1-norm than the composed Hamiltonian with PK classically partitioned (Track BF: electronic-only 1-norm 32.6 Ha). The sparsity advantage in Pauli count may be partially offset by a 1-norm disadvantage. This tradeoff must be quantified for the balanced Hamiltonian.

The mathematical infrastructure for computing the missing integrals exists in the Coulomb Sturmian literature and connects directly to GeoVac's S^3 foundation. The open question is convergence: does the Shibuya–Wulfman expansion converge fast enough at molecular bond lengths to preserve the structural sparsity that is GeoVac's primary computational advantage?

If the answer is yes, the composed framework would be superseded by coupled composition: same block structure, same Gaunt selection rules, the composed builder's structural sparsity retained (a Pauli count exactly linear in Q across molecules at fixed basis, Paper 14), but without PK and with significantly better accuracy. Paper 17's composed architecture would then move to the archive, replaced by the coupled composition as the production molecular Hamiltonian builder.

If the answer is no, the composed framework with PK remains the production architecture, and the accuracy ceiling is accepted as the structural cost of maintaining block-diagonal sparsity. In either case, this paper documents the mathematical connections and the concrete data (Track CB census, FCI benchmarks) that define the research frontier.

The $n_{\max} = 3$ results with analytical integrals definitively resolve the key open questions from Phase 3. The energy convergence is excellent through $n_{\max} = 3$: 1.8% $\to$ 0.20% (9× improvement) as the basis grows from $Q = 30$ to $Q = 84$, with the variational bound respected at these basis sizes (at $n_{\max} = 4$ the model crosses past exact; the well-shape subsection). This rules out the PK-like divergence scenario (where higher angular momentum channels systematically overcounted correlation).

The R_{eq} drift, however, is confirmed as structural to the block decomposition, not numerical. Analytical integrals (machine precision via incomplete gamma functions) changed R_{eq} by only 0.007 bohr at $n_{\max} = 3$ (3.287 $\to$ 3.280 bohr), while the drift from $n_{\max} = 2$ to 3 is +0.053 bohr. This drift is 3× smaller than PK's +0.15–0.22 bohr/$l_{\max}$ but in the same outward direction. The root cause is the block decomposition itself: each block uses a different effective nuclear charge (Z_{eff}), and orbitals on different blocks do not span the same Hilbert space. The cross-center V_{ne} partially bridges this gap but cannot fully compensate for the incomplete orbital overlap between blocks at finite basis. The balanced approach is optimal for single-point quantum simulation at fixed geometries (0.20% energy error), but not for PES-based equilibrium geometry determination. The analytical evaluation of V_{ne} via incomplete gamma functions adds the cross-center nuclear attraction to the algebraic registry (Paper 18, Table I), confirming that the balanced coupled framework introduces no new exchange constants relative to the composed architecture.

Relationship to the LCAO negative result. The coupled composition program reconstructs a multi-center molecular Hamiltonian from atom-centered orbitals, which is superficially similar to the LCAO approach abandoned in v0.9.x (Section 3 of the project guidelines). The key difference: LCAO used unstructured graph concatenation with no natural geometry, producing R-independent kinetic energies. Coupled composition retains the natural geometry block structure—each electron group inhabits its own optimal coordinate system—and Gaunt selection rules enforce sparsity within and across blocks. The block decomposition provides physical structure that LCAO lacked.

Applications paper. Paper 20 presents the GeoVac Hamiltonian library as a practical tool for the quantum computing community, with benchmark tables comparing GeoVac and Gaussian resource costs at matched accuracy, FCI PES characterization, and installation instructions. The balanced coupled architecture documented in this paper (Paper 19) provides the PK-free Hamiltonians benchmarked there.

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